

Monday: Prior Knowledge1. Domain and range of $f(x) = \sqrt{x}$?

D: _____ R: _____

2. If $f(x) = x^2$, find:

- $f(-3) =$ _____
- $f(0) =$ _____
- $f(2) =$ _____

3. [SAT] Equation parallel to $y = 2x - 5$?

- A) $y = -2x + 3$
- B) $y = 2x + 7$
- C) $y = \frac{1}{2}x - 5$
- D) $y = -\frac{1}{2}x + 1$

4. Solve: $\sqrt[3]{2x+3} = 5$ **Tuesday: Parent Functions**1. $f(x) = x^3$ Table:

x	-2	-1	0	1	2
y					

Domain: _____ Range: _____

2. $g(x) = \sqrt[3]{x}$ Table:

x	-8	-1	0	1	8
y					

3. Symmetry type for both?

4. [SAT] Solve $3(x - 2) > 5x + 4$.

- A) $x < -5$
- B) $x > -5$
- C) $x < 5$
- D) $x > 5$

Wednesday: Writing Equations1. $f(x) = \sqrt[3]{x}$ shifted **right 5** and **down 3**. $g(x) =$ _____2. $f(x) = x^3$ **reflected over x-axis** and shifted **left 2**. $g(x) =$ _____

3. Identify type & transformations:

$$f(x) = -2(x + 4)^3 - 7$$

Type: _____

Transformations: _____

4. [SAT] If $3x + 2y = 12$, find y when $x = -2$.

- A) 3
- B) 6
- C) 9
- D) 12

Thursday: Cubic Transformations

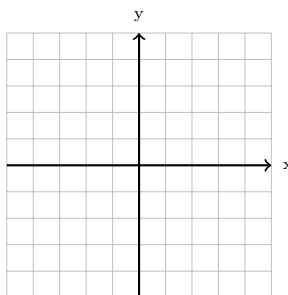
1. Describe
- all**
- transformations:

$$g(x) = -(x + 2)^3 - 1$$

2. Solve:
- $(x + 5)^{1/3} = 3$

3. Graph

$$f(x) = 2(x - 1)^3.$$



4. [SAT]
- $y = -3x + 7$
- is
- $\perp$
- to:

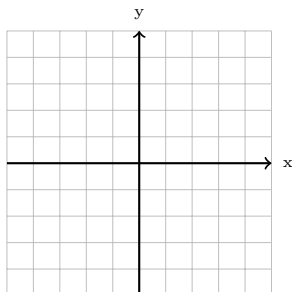
- A) $y = -3x + 2$
 B) $y = 3x - 1$
 C) $y = \frac{1}{3}x + 4$
 D) $y = -\frac{1}{3}x + 5$

Friday: Cube Root Transformations

1. Transformation for
- $g(x) = \sqrt[3]{x} - 4$
- :

2. Transformation for
- $g(x) = \sqrt[3]{x + 3}$
- :

3. Graph
- $f(x) = \sqrt[3]{x - 2} + 1$
- .



4. [SAT] Line
- m
- slope =
- $\frac{3}{4}$
- . Slope of
- $\perp$
- line?

- A) $\frac{3}{4}$
 B) $-\frac{3}{4}$
 C) $\frac{4}{3}$
 D) $-\frac{4}{3}$